Personal Statement

PS · MASTER'S APPLICATION

APPLICATION PROFILE / 学生与申请信息

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本科院校	北京大学 / Peking University	**本科专业**	计算机科学 / Computer Science
本科 GPA	3.8/4.0	**语言成绩**	IELTS 7.5 · TOEFL 105 · GRE 330
申请院校	卡内基梅隆大学 / CMU	**申请专业**	人工智能 / MSAII
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1. Motivation and Defining Experience

My interest in artificial intelligence became concrete while I was working on a national undergraduate innovation project titled “Deep Learning-Based Medical Image Analysis System.” The project exposed me to noisy real-world data, limited annotations, and the difficulty of aligning images with clinical text. I learned that an improvement on a conventional validation set does not automatically translate into reliable performance in a complex clinical setting. That experience changed the way I viewed AI: meaningful progress requires not only a higher score, but also methods that can be explained, tested, and adapted to imperfect environments. It motivated me to pursue graduate training in robust multimodal learning.

2. Research Interests

I am particularly interested in robust multimodal representation learning under limited-data conditions. My current questions concern semantic alignment between images and text, the stability of contrastive objectives, and the role of attention mechanisms in cross-modal fusion. I want to investigate how models can generalize when training samples are scarce and class distributions are imbalanced. Rather than treating these issues as abstract benchmarks, I hope to begin with problems observed in real projects, formulate testable hypotheses, and evaluate each proposed method through reproducible experiments and careful error analysis.

3. Academic Preparation

As a Computer Science undergraduate at Peking University, I have built a systematic foundation in data structures and algorithms, operating systems, machine learning, deep learning, and natural language processing. My GPA of 3.8/4.0 reflects consistent academic performance and disciplined study. These courses taught me to move from problem formulation to implementation and experimental evaluation. My GRE score of 330, IELTS score of 7.5, and

TOEFL score of 105 have also prepared me to study, write, and collaborate in an English-speaking academic environment.

4. Research, Internship, and Project Experience

In the medical imaging project, I contributed to model design and core experiments, with particular attention to performance bottlenecks in cross-modal alignment under small-sample conditions. During my NLP internship at Baidu AI Lab, I worked on intent classification and slot filling for a dialogue system. Through iterative feature and model improvements, the intent-classification accuracy increased from 91.2% to 94.6%. I also placed in the top 5% of a Kaggle multimodal classification challenge. Together, these experiences taught me to define a measurable problem, implement a solution, document the process, and connect technical choices to verifiable outcomes.

5. Why Carnegie Mellon and MSAII

Carnegie Mellon University's M.S. in Artificial Intelligence and Innovation is a strong fit for my goal of combining rigorous technical training with real-world problem solving. The broader School of Computer Science environment and the resources of the Machine Learning Department would allow me to deepen my understanding of robust learning, multimodal reasoning, and large-scale model development. I am also drawn to the program's emphasis on translating advanced AI methods into usable systems. This combination would help me turn my existing project experience into a more disciplined capacity for research and innovation.

6. Future Goals

In the short term, I hope to strengthen my theoretical foundation, participate in reproducible research, and become more precise in defining questions, designing experiments, and interpreting results. In the longer term, I plan to work at the intersection of AI research and engineering, with a focus on deploying reliable multimodal systems in high-impact domains such as healthcare. I believe that CMU's academic community would challenge me to grow beyond my current experience while giving me the tools to contribute responsibly to the development of practical artificial intelligence.